

Databricks DE Associate — 45- Question Practice Test

Exam-style questions with a full answer key + explanations

90 minutes · all 5 domains · passing $\approx$ 32/45

Exam-style multiple choice. Give yourself **90 minutes**, answer all 45, then check the **Answer Key** at the end. Passing $\approx 70\%$ ($\approx 32/45$). Questions are grouped by domain but mixed in difficulty.

Questions

1. The Databricks Lakehouse combines: A) OLTP + NoSQL B) Data lake storage + data warehouse reliability/performance C) Only Spark D) Only SQL
2. In the Databricks architecture, where do the clusters (compute) that process your data run? A) Databricks control plane B) Your cloud account (data plane) C) On your laptop D) In DBFS
3. You drop an **external** table (created with `LOCATION`). What happens to the files? A) Deleted B) Kept in the storage location C) Moved to trash D) Corrupted
4. Delta Lake adds which capability on top of Parquet? A) Only compression B) ACID transactions, time travel, schema enforcement C) OLTP indexing D) Nothing
5. Which cluster type is most cost-effective for a scheduled batch job? A) All-purpose cluster B) SQL warehouse C) Job cluster D) Cluster pool only
6. What is DBFS? A) A database engine B) A filesystem abstraction over cloud object storage C) A cluster type D) A governance tool
7. Which best reduces cluster cost from idle clusters? A) Larger nodes B) Auto-termination C) More workers D) Caching
8. A **global temp view** is accessible: A) Only in the current notebook B) Across all notebooks on the same cluster C) Across all workspaces D) Forever
9. Which command creates a table from a query result? A) `INSERT INTO` B) `COPY INTO` C) `CREATE TABLE ... AS SELECT` D) `MERGE`
10. To read a JSON file directly with Spark SQL: A) ``SELECT * FROM json.`/path/`` B) `LOAD json` C) `IMPORT json` D) `READ json``
11. Repos in Databricks provide: A) Cluster management B) Git version control integration C) Secret storage D) SQL dashboards
12. Which write mode replaces all existing data in a Delta table? A) `append` B) `overwrite` C) `ignore` D) `merge`
13. You must upsert (update matching rows, insert new ones) into a Delta table. Use: A) `INSERT INTO` B) `MERGE INTO` C) `COPY INTO` D) `UPDATE`
14. `COPY INTO` is best described as: A) A streaming source B) An idempotent, retrievable bulk file load into a Delta table C) A cluster policy D) A view type
15. Auto Loader (`cloudFiles`) is used to: A) Compact files B) Incrementally ingest new files as a stream with checkpointing C) Grant permissions D) Schedule jobs

16. In Structured Streaming, exactly-once processing requires: A) `cache()` B) A checkpoint location C) `OPTIMIZE` D) `overwrite` mode
17. `OPTIMIZE ... ZORDER BY (col)` does what? A) Deletes old files B) Compacts small files and co-locates data for skipping C) Grants access D) Creates a view
18. `VACUUM` on a Delta table: A) Compacts files B) Removes data files no longer referenced (default 7-day retention) C) Rebuilds indexes D) Refreshes stats
19. Query a Delta table as of version 5: A) `SELECT FROM t@5` B) `SELECT FROM t VERSION AS OF 5` C) `ROLLBACK 5` D) `RESTORE 5`
20. Which is NOT a Delta Lake benefit? A) ACID B) Time travel C) Schema enforcement D) Sub-millisecond single-row OLTP lookups
21. To avoid shuffling a large table when joining a small dimension: A) `repartition` B) `coalesce` C) Broadcast the small table D) `cache` the large table
22. A transformation in Spark is: A) Executed immediately B) Lazy — builds the plan; runs on an action C) Always a shuffle D) A cluster setting
23. Which is an **action** (triggers execution)? A) `filter` B) `select` C) `count` D) `withColumn`
24. Deduplicate keeping the latest row per key — best approach: A) `DISTINCT` B) `ROW_NUMBER()` `OVER (PARTITION BY key ORDER BY ts DESC) = 1` C) `GROUP BY key` D) `LIMIT 1`
25. A temp view created with `CREATE OR REPLACE TEMP VIEW` is: A) Permanent B) Session-scoped C) Cluster-scoped D) Cross-workspace
26. `INSERT OVERWRITE` on a partitioned table (dynamic) affects: A) All partitions always B) Only the partitions present in the new data (with dynamic overwrite) C) No partitions D) Metadata only
27. To flatten an array column into rows: A) `split()` B) `explode()` C) `collect_list()` D) `flatten()` only
28. Extract a field from a JSON string column `payload`: A) `payload->field` B) `payload:field` / `from_json` C) `payload.field()` D) `get(payload)`
29. Which statement about **managed** tables is true? A) Dropping keeps the data B) Dropping deletes both metadata and data C) You must specify LOCATION D) They can't be Delta
30. Delta Live Tables (DLT) `@dlt.expect_or_drop("r", "amount>0")` will: A) Fail the pipeline on violation B) Drop rows violating the rule, keep running C) Only log a warning D) Ignore the rule
31. DLT `@dlt.expect_or_fail(...)` on violation: A) Drops rows B) Warns only C) Fails the pipeline/update D) Auto-fixes data
32. A Databricks **Job** can: A) Only run one task B) Chain multiple tasks with dependencies, schedule, and retries C) Not use clusters D) Only run SQL
33. Best way to alert on a failed scheduled job: A) Poll manually B) Configure email/webhook notifications on failure C) Cache results D) Increase workers

34. In DLT, a `STREAMING LIVE TABLE` is used for: A) Static lookups B) Incrementally processed streaming sources C) Governance D) Cluster config
35. Unity Catalog's namespace is: A) `schema.table` B) `catalog.schema.table` C) `database.table` D) `table` only
36. Grant read on a table to a group in Unity Catalog: A) `GRANT SELECT ON TABLE cat.sch.t TO group` B) `ALLOW READ` C) `USE TABLE` D) `SHOW GRANTS`
37. Column masking / row filtering for non-privileged users is typically done with: A) A cluster policy B) A dynamic view (`is_account_group_member`) C) `VACUUM` D) A secret scope
38. Where should database passwords be stored? A) In the notebook code B) In a Databricks secret scope, read via `dbutils.secrets.get` C) In DBFS as text D) In a table
39. Which reduces Spark job time by skipping data at read? A) `collect()` B) Partition pruning + ZORDER data skipping C) More UDFs D) `orderBy`
40. A cluster **policy** is used to: A) Grant table access B) Constrain cluster config (size, node type, auto-termination) for standardization/cost C) Schedule jobs D) Store secrets
41. Which format does Delta store data files in? A) CSV B) Parquet C) Avro D) ORC
42. The Delta transaction log directory is: A) `/_checkpoints` B) `/_delta_log` C) `/_metadata` D) `/_history`
43. `DESCRIBE HISTORY table` shows: A) The schema B) The version/operation history (for time travel/audit) C) Row counts only D) Grants
44. To incrementally load only new files repeatedly and idempotently in batch, prefer: A) `spark.read` full reload each run B) `COPY INTO` (or Auto Loader) C) `DISTINCT` D) `OPTIMIZE`
45. Which is true about SQL Warehouses? A) They run notebooks B) They provide SQL compute for BI/dashboards (serverless/pro/classic) C) They are governance objects D) They replace Delta
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ANSWER KEY (with one-line why)

1. **B** — lakehouse = lake storage + warehouse reliability.
2. **B** — clusters run in the customer's data plane; control plane is managed by Databricks.
3. **B** — external tables keep files on drop; managed delete data.
4. **B** — Delta adds ACID, time travel, schema enforcement.
5. **C** — job clusters spin up per run, cheapest for scheduled work.
6. **B** — DBFS is a filesystem over cloud storage.
7. **B** — auto-termination stops idle clusters.
8. **B** — global temp view = cluster-scoped; temp view = session.
9. **C** — CTAS creates a table from a query.
10. **A** — file-source SQL: ``json`/path``.

11. **B** — Repos = Git integration.
12. **B** — `overwrite` replaces data.
13. **B** — MERGE = upsert.
14. **B** — COPY INTO = idempotent bulk file load.
15. **B** — Auto Loader = incremental streaming file ingestion.
16. **B** — checkpoint gives exactly-once.
17. **B** — OPTIMIZE compacts; ZORDER co-locates for skipping.
18. **B** — VACUUM removes unreferenced files (7-day default).
19. **B** — `VERSION AS OF`.
20. **D** — Delta is analytics, not OLTP.
21. **C** — broadcast the small table.
22. **B** — transformations are lazy.
23. **C** — `count` is an action.
24. **B** — ROW_NUMBER window dedup.
25. **B** — temp view is session-scoped.
26. **B** — dynamic partition overwrite affects only present partitions.
27. **B** — `explode()` array → rows.
28. **B** — `payload:field` / `from_json`.
29. **B** — managed drop deletes data.
30. **B** — expect_or_drop drops bad rows, keeps running.
31. **C** — expect_or_fail fails the pipeline.
32. **B** — Jobs chain tasks with deps/schedule/retries.
33. **B** — configure failure notifications.
34. **B** — streaming live table = incremental streaming source.
35. **B** — three-level `catalog.schema.table`.
36. **A** — `GRANT SELECT ON TABLE ... TO group`.
37. **B** — dynamic view with group checks.
38. **B** — secret scope + `dbutils.secrets.get`.
39. **B** — pruning + ZORDER skip data.
40. **B** — cluster policy constrains config/cost.
41. **B** — Delta stores Parquet files.
42. **B** — `_delta_log`.
43. **B** — DESCRIBE HISTORY = version/operation history.
44. **B** — COPY INTO / Auto Loader for idempotent incremental loads.
45. **B** — SQL Warehouses = SQL compute for BI.

Score: /45. ~32+ = **passing range**. Review every miss and re-read that topic in the prep sheet.

